Differentiation: Stationary Points

Questions 1–13

Question 1. For each of following, find the value(s) of x for which $\dfrac{dy}{dx} = 0$.

a) $y = x^2 - 2x$

b) $y = -2x^2 + 18x - 37$

c) $y = x^3 - 3x^2 + 3x - 4$

d) $y = x - \dfrac{4}{x^2}$

e) $y = x + \dfrac{4}{x}$

f) $y = \dfrac{x^3 + 2}{x}$

g) $y = \dfrac{x^4 + 3}{3x^3}$

Question 2. Find the interval on which each of the following functions are decreasing, giving your answer in set notation:

a) $f(x) = x^2 + 4x$

b) $f(x) = 6x^2 - 10x + 5$

c) $f(x) = x^3 - 3x$

d) $f(x) = 2x^3 + 3x^2 - 36x - 10$

e) $f(x) = -2x^3 - 18x^2 - 48x - 1$

Question 3. Find the interval on which the function

$$f(x) = -x^3 - 6x^2 + 15x + 1$$

is increasing, giving your answer in interval notation.

Question 4. Show that the function

$$f(x) = x^3 - 3x^2 + 4x - 8$$

is always increasing.

Question 5. Show that the function

$$f(x) = -3x^3 + 3x^2 - 8x + 2$$

is always decreasing.

Question 6. Find the coordinates of the stationary points of the following curves and determine their types:

a) $y = -x^2 + 5x - 6$

b) $y = 2x^3 - 15x^2 + 24x$

c) $y = 2x^3 - 3x^2 - 12x + 7$

d) $y = -x^3 + 3x^2 + 72x$

e) $y = 2x^4 - 4x^2$

f) $y = 4x + \dfrac{1}{x}$

g) $y = \dfrac{1}{x^2} + 2x$

h) $y = 3x^6 - 6x^3$

Question 7. Find the coordinates of the maximum of the gradient function of

$$y = -3x^3 - 2x^2 + 1.$$

Question 8. Find the coordinates of the stationary points, determine their types and hence sketch each of the following curves:

a) $y = x^3 - 3x^2 - 45x$

b) $y = 16x^4 - 32x^2 + 7$

c) $y = 5x - 2x^{\frac{1}{2}}$ for $x \geq 0$

Question 9. Find the values of a, b and c such that

$$y = -x^3 + ax^2 + bx + c$$

has stationary points when $x = 1$ and $x = 3$, and passes through the point $(2, 4)$.

Question 10. Find the values of a, b and c such that

$$y = ax^4 + bx^2 + c$$

passes through the points $(1, 0)$ and $(0, -3)$, and has a stationary point at $(\sqrt{2}, 1)$. Find all the stationary points and determine their types.

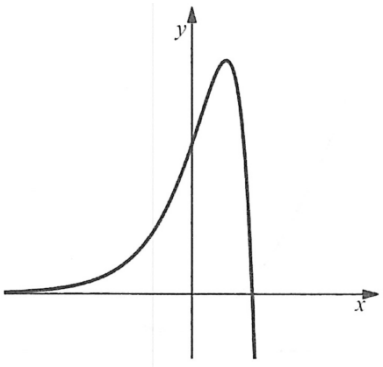
Question 11.

Prerequisite knowledge for Q11–Q13: Differentiation of the form ae^{kx}

Giving your answer in set notation, find the interval on which $f(x) = e^{2x} - x$ is increasing.

Question 12. The diagram shows the curve

$$y = 5e^x - e^{2x}.$$



a) Find the coordinates at which the curve crosses the x-axis and the y-axis.

b) Find the coordinates of the stationary point, giving your answer in exact form.

c) Show that the stationary point is a maximum.

Question 13. Find the coordinates of the stationary point of

$$y = e^{3x} - e^{2x} - e^{x}$$

and determine its type.